

Juan José Restrepo Rosero

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PROFILE

Trilingual Electronics Engineer with a major in Mechatronics and a minor in Computer Science from the Javeriana University in Cali, Colombia. I specialize in Software Development, Machine Learning, Robotics and Automation. I am a committed, responsible, dedicated and persistent person. Great ability with numbers, multicultural teamwork and motivation to learn new things and approach challenges with positive and creative thinking.

PROFESSIONAL EXPERIENCE

Veevart – DevOps Engineer: December 2024 – May 2025

- Developed and maintained backend services in **Java** and **Apex** within a production environment, integrated with **Salesforce**, ensuring scalability, reliability, and maintainability of core application logic.
- Collaborated on **backend architecture** and **debugging processes**, identifying performance issues, improving error handling, and supporting stable integrations between services and external platforms.
- Implemented automation workflows for development and deployment processes using **Python** and **Bash**, improving operational efficiency and reducing manual intervention.
- Managed application **releases** and **version control**, applying **static code analysis (PMD)** and enforcing coding standards, contributing to improved code quality, maintainability, and production reliability.

Holcim ABS - Electronic Engineering Intern: July 2023 - January 2024

- Designed, developed, and deployed **RPA automations** with **UiPath**, integrated with **SAP** and **BTP** services, automating key transactional and reporting processes.
- Built scalable automation workflows using **VBA Macros** and **Google Apps Script**, reducing manual workload and improving data accuracy.
- Collaborated with cross-functional teams to identify automation opportunities, standardize documentation, and ensure continuous improvement.

EDUCATION

Pontificia Universidad Javeriana Cali,

Department of Electronics and Computer Science.

Master's degree in data science, July 2024 - In Progress

Santiago de Cali, CO

Department of Electronics and Computer Science.

Bachelor of Science in Electronics Engineering, January 2019 -April 2024

Cumulative GPA: 3.69/4.00

Southern Lee High School

Foreign Exchange Student, High School Diploma, August 2017 -June 2018

Cumulative GPA: 3.70/4.00

Sanford, NC, USA

San Antonio María Claret

High School Student, September 2011 -June 2017

Cumulative GPA: 3.80/4.00

Santiago de Cali, CO

SKILLS

Programming & Scripting:

Python, SQL, Java/Apex, JavaScript/TypeScript, C/C++, Bash

Backend Development:

Node.js (Express), RESTful APIs (CRUD), asynchronous programming, API deployment, request validation, error handling

Software Engineering & Code Quality:
Static Code Analysis (PMD; Ruff; mypy); Linting & Formatting,
modular architecture, maintainable system design

Data Science & Machine Learning:
scikit-learn, pandas, NumPy, TensorFlow/Keras, NLP (spaCy,
NLTK), data modeling, feature engineering, statistical analysis, R

Tools & Frameworks:
Node.js; Neo4j; MATLAB/Simulink

Data Engineering & DevOps:
Apache Airflow; dbt; Docker; ETL/ELT pipelines;
PostgreSQL; MySQL; Linux; Git

Automation & Scripting:
UiPath; VBA Macros; Google Apps Script

Languages:
Spanish (Native); English (C1 TOEFL iBT)
German (B1 Goethe); French (A2)

PROJECTS

Master's Thesis – Binary MIL for WSI Classification: May 2025 – Jan 2026

- Developed a binary classification pipeline for histopathological Whole Slide Images (WSI) using a Multiple Instance Learning (MIL) approach, treating each slide as a bag of image patches with weak slide-level labels. The workflow includes patch extraction, deep feature embedding, and attention-based aggregation to produce slide-level predictions, with patient-level cross-validation and standard classification metrics.

Knowledge Management & Spectroscopy Data Pipeline: May 2025 – January 2026

- Developed an ETL/ELT pipeline for spectroscopy data using Airflow, PostgreSQL, and Python, following a Medallion Architecture to extract, clean, transform, and model experimental data. The project consolidates tacit knowledge and dispersed research data from multiple techniques (Raman, FTIR, UV-Vis, and others), applies data engineering to detect patterns and trends, and generates analytical outputs to support research decision-making and services for the scientific community of the iOmicas Institute.

Qversity Data Engineering Project: June 2025 – July 2025

- Developed an ELT pipeline using Apache Airflow for orchestration, dbt for transformations, and PostgreSQL as the data warehouse, implementing Medallion Architecture to extract, clean, transform, and model data from AWS S3, while leveraging Docker to streamline environment setup and pipeline deployment. Delivered 21 actionable business insights including ARPU by plan, revenue by geography, customer segmentation, and payment behavior, ensuring data quality through testing, validation, and structured data pipelines, incorporating best practices for reproducibility, error handling, and maintainability.

Tasks Manager API: March 2026

- Developed a RESTful API (CRUD) for task management using Node.js and Express, implementing asynchronous request handling, middleware architecture, and structured routing. Deployed the application in a production environment using Render, managing environment variables and ensuring reliable API availability. Applied backend engineering best practices including input validation, error handling, and modular code organization to improve maintainability and system robustness.

World Cup Prediction System: March 2026

- Developed a data-driven prediction system integrating data collection, feature engineering, and machine learning models to forecast match outcomes. Structured the pipeline following reproducible data science practices, including data validation, preprocessing, and model evaluation, ensuring consistency and reliability in the modeling workflow.

Formula 1 Predictor: March 2025

- Developed a data science application to forecast the 2025 Formula 1 season outcomes by integrating web-scraped historical race data with advanced machine learning models. The project involved thorough data cleaning, feature engineering to extract key performance metrics for drivers and teams and evaluating models such as logistic regression and random forests.

REFERENCES

Carol Villar

Director Network Engineering

Claro Enterprise Solutions

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